

Tournament Predictions

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In [1]: #setup
from rugbyModels import *
np.random.seed(1)
import warnings
warnings.filterwarnings("ignore", category=FutureWarning)

data= pd.read_csv("final_data_1.csv")
data["date"] = pd.to_datetime(data["date"])

#think its reasoable to combine these for models without dgs - shouldnt overwrit
data["home_penalties"] = data["home_penalties"]+data["home_drop_goals"]
data["away_penalties"] = data["away_penalties"]+data["away_drop_goals"]
```

```
In [2]: ncPred = PoissonModel(nSamples = 20000).fit(data) # fit model
NHtable,SHtable = ncPred.simulate_group() # simulate group stage
```

Initializing NUTS using jitter+adapt_diag...
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [home, sd_att, sd_def, att, def_, int_try, sd_dis_clean, sd_dis_force, dis_clean, dis_force, int_pen, p_conv]
Output()

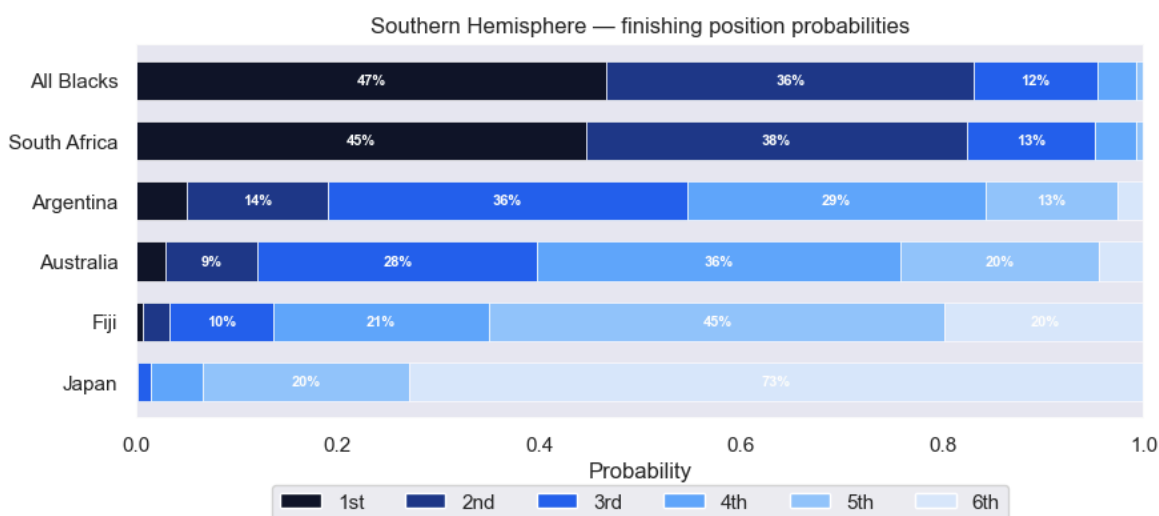
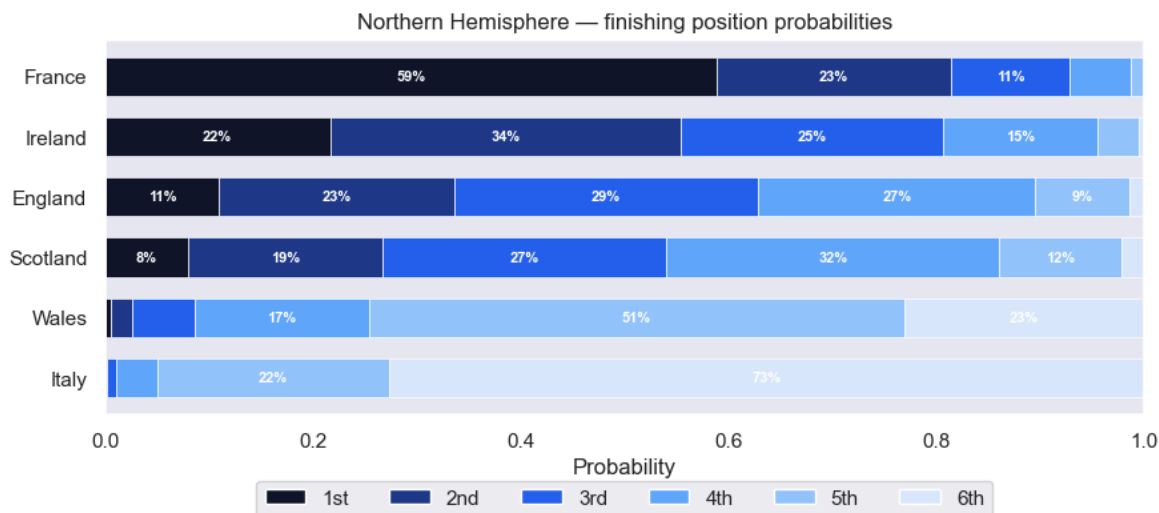
Sampling 4 chains for 1_500 tune and 5_000 draw iterations (6_000 + 20_000 draws total) took 90 seconds.

=== Northern Hemisphere ===

	mean_pts	p_1st	p_top2	p_top3	p_top4	p_top5
France	23.467	0.588	0.814	0.927	0.986	0.999
Ireland	19.828	0.217	0.554	0.807	0.955	0.995
England	17.766	0.110	0.336	0.629	0.895	0.986
Scotland	16.948	0.080	0.267	0.540	0.860	0.978
Wales	10.778	0.006	0.026	0.086	0.254	0.769
Italy	6.432	0.000	0.002	0.011	0.050	0.274

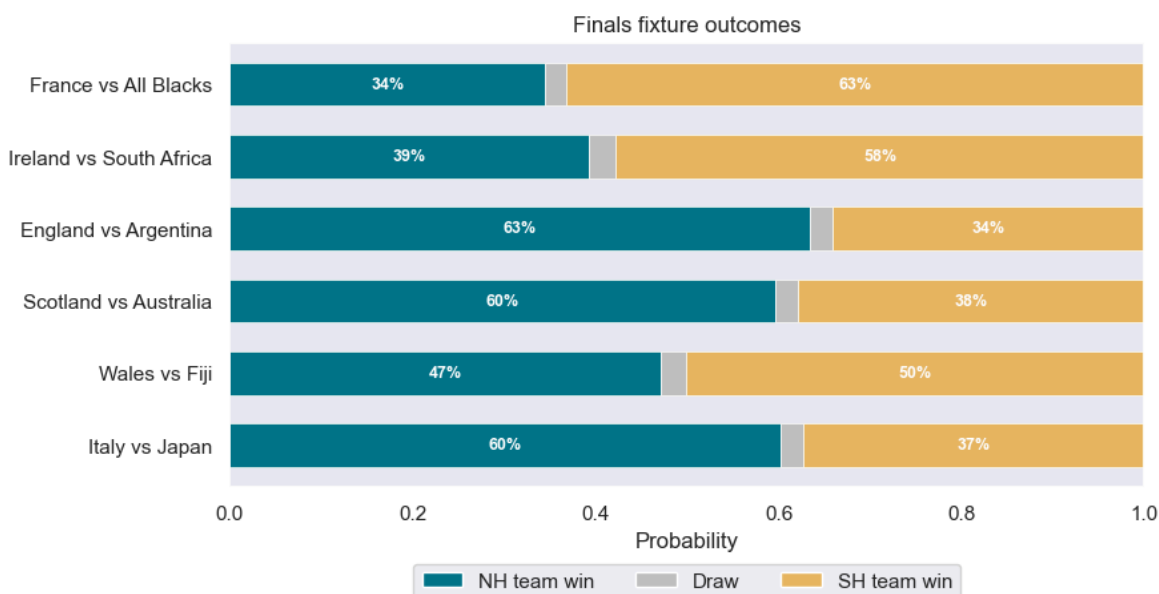
=== Southern Hemisphere ===

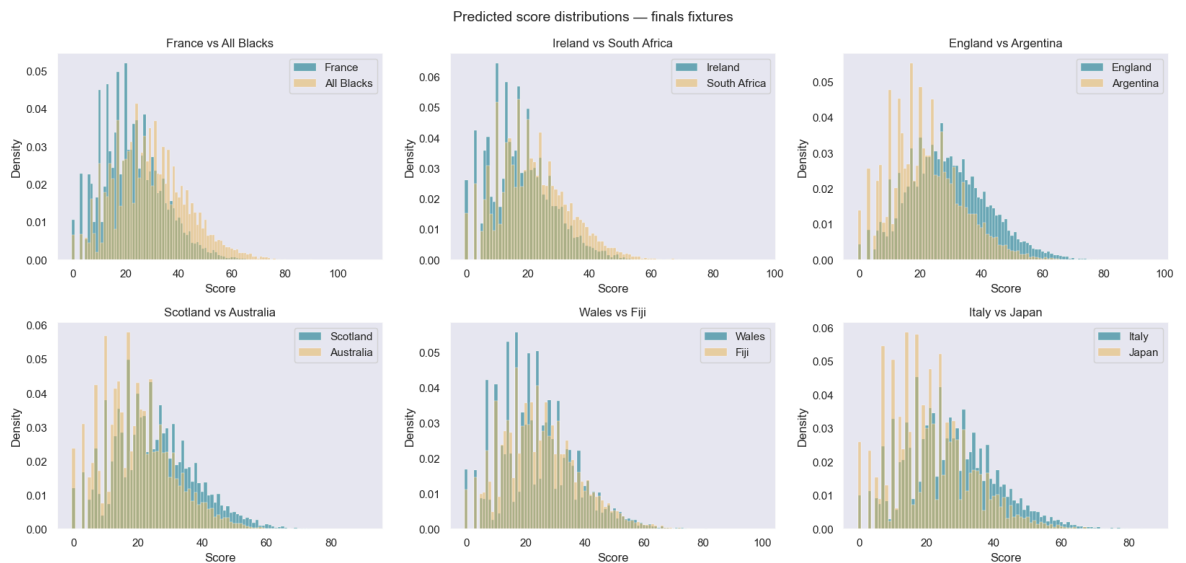
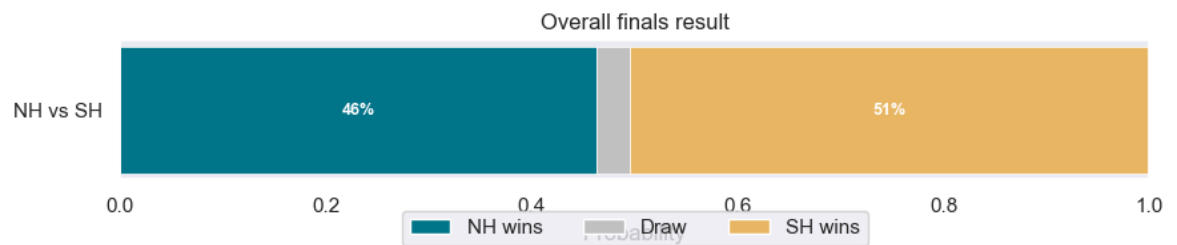
	mean_pts	p_1st	p_top2	p_top3	p_top4	p_top5
All Blacks	21.934	0.467	0.830	0.954	0.992	1.000
South Africa	21.787	0.447	0.824	0.950	0.992	0.999
Argentina	14.844	0.050	0.190	0.547	0.842	0.973
Australia	13.373	0.029	0.121	0.398	0.757	0.955
Fiji	9.231	0.007	0.033	0.136	0.351	0.802
Japan	4.592	0.000	0.002	0.015	0.067	0.271



And if we assume these group placings, how will the finals go?

```
In [3]: _ = ncPred.simulate_finals(NHtable, SHtable)
```





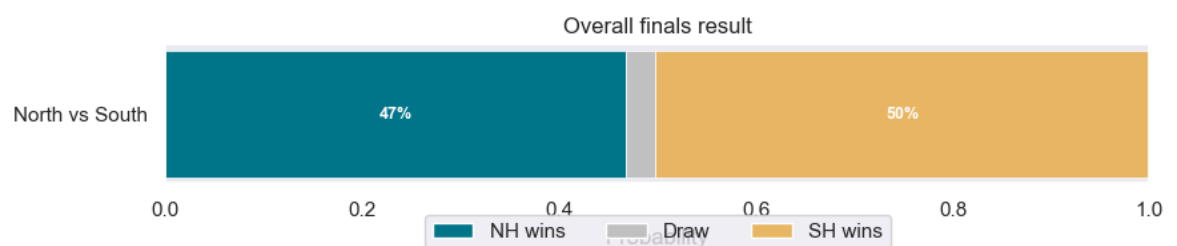
=== Finals fixtures ===

France vs All Blacks: $P(\text{North})=0.345$ $P(\text{Draw})=0.023$ $P(\text{South})=0.632$
 Ireland vs South Africa: $P(\text{North})=0.393$ $P(\text{Draw})=0.029$ $P(\text{South})=0.578$
 England vs Argentina: $P(\text{North})=0.635$ $P(\text{Draw})=0.024$ $P(\text{South})=0.341$
 Scotland vs Australia: $P(\text{North})=0.597$ $P(\text{Draw})=0.025$ $P(\text{South})=0.379$
 Wales vs Fiji: $P(\text{North})=0.472$ $P(\text{Draw})=0.027$ $P(\text{South})=0.501$
 Italy vs Japan: $P(\text{North})=0.602$ $P(\text{Draw})=0.025$ $P(\text{South})=0.374$

Overall: $P(\text{NH wins}) = 0.463$ $P(\text{Tie}) = 0.032$ $P(\text{SH wins}) = 0.505$

And if we don't assume these placings and factor uncertainty over group stage positions into finals calculations?

```
In [4]: _ = ncPred.simulate_finals_from_scratch()
```



=== Finals ===

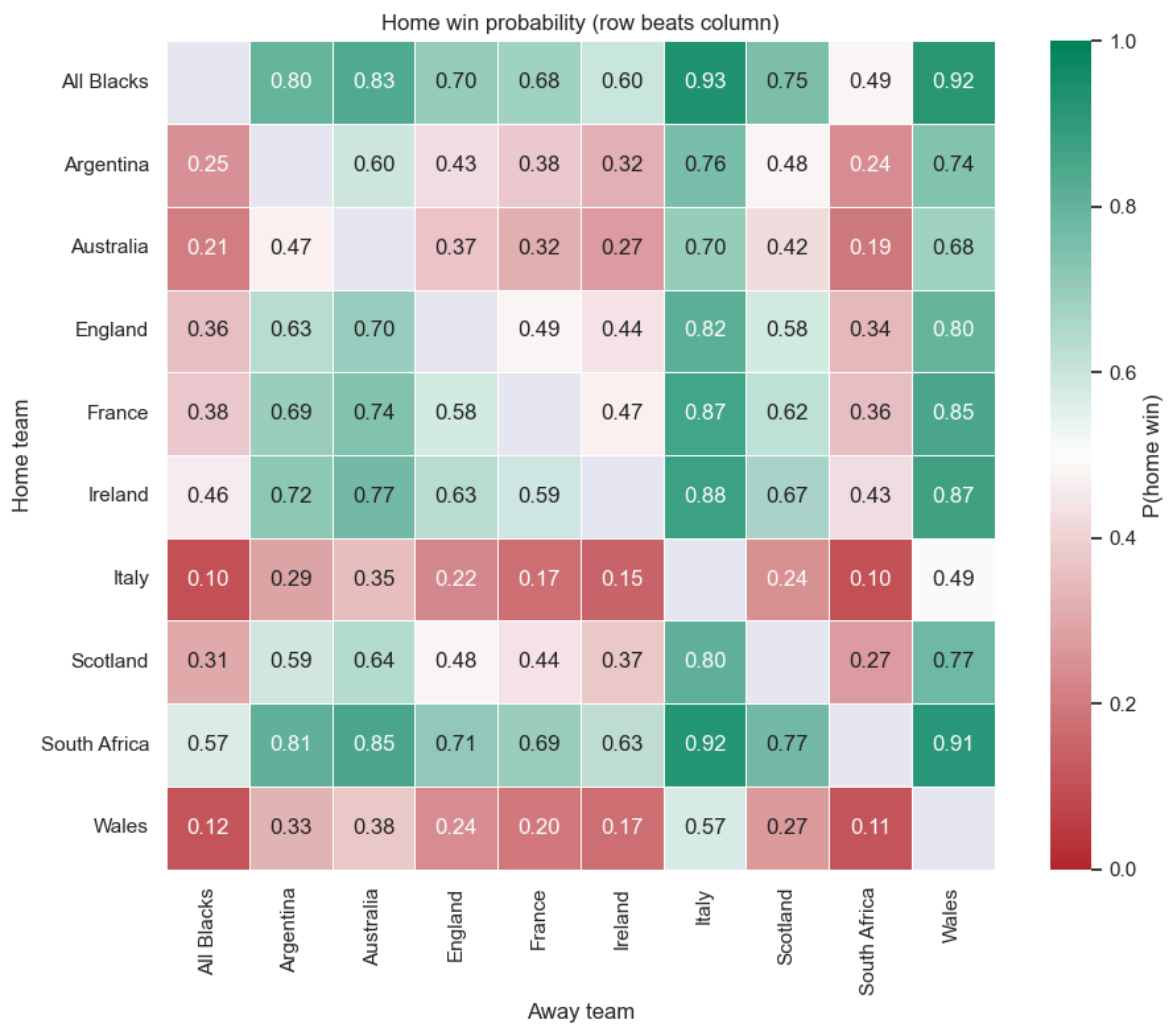
Overall: $P(\text{North}) = 0.468$ $P(\text{Tie}) = 0.030$ $P(\text{South}) = 0.502$

Looking close!

Pairwise win probabilities

The probability of each team winning a home fixture against each other team.

```
In [5]: ncPred.plot_pairwise_win_probs()
```



```
Out[5]:
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	All Blacks	Argentina	Australia	England	France	Ireland	Italy	Scotland
All Blacks	NaN	0.79845	0.83330	0.70300	0.67625	0.60105	0.93330	0.74840
Argentina	0.24950	NaN	0.59710	0.42925	0.37895	0.32115	0.75965	0.47680
Australia	0.20635	0.46575	NaN	0.36930	0.32010	0.27115	0.70250	0.41905
England	0.35700	0.63315	0.69550	NaN	0.48555	0.43760	0.81935	0.57920
France	0.38245	0.68810	0.74080	0.58155	NaN	0.46815	0.86670	0.62165
Ireland	0.46345	0.72480	0.76810	0.62985	0.59220	NaN	0.88075	0.66630
Italy	0.09620	0.29125	0.34945	0.22305	0.17060	0.14800	NaN	0.24255
Scotland	0.30535	0.58900	0.64445	0.47850	0.44100	0.37275	0.80495	NaN
South Africa	0.56615	0.80765	0.84930	0.71445	0.69435	0.62655	0.92315	0.76685
Wales	0.11610	0.32620	0.37845	0.24055	0.19885	0.16700	0.57085	0.26675

Interpretability

What is driving each teams success/failure?

```
In [6]: ncPred.plot_params()
```

